Dysfunctional growth hormone receptor in a strain of sex-linked dwarf chicken: evidence for a mutation in the intracellular domain

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Abstract

The sex-linked dwarf (*dwdw*) chicken represents a valuable animal model for studying GH insensitivity and the consequence of mutations in the GH receptor (GHR) gene. We have recently reported undetectable hepatic GH-binding activity and an aberrantly sized transcript in a strain of *dwdw* chickens obtained from Arbor Acre Farms, Inc. (Glastonbury, CT, USA). Southern blot analysis of the chicken GHR (cGHR) gene revealed a restriction-fragment length polymorphism in HindIII and EcoRI digests of genomic DNA in this strain of *dwdw* chicken. In order to localize the molecular mutation, we analysed the gene structure and determined the complete sequence of the 3′ untranslated region (3′ UTR) of the normal cGHR. With the use of this information, we located a large deletion in the 3′ end of the cGHR gene of the Connecticut (CT) strain of *dwdw* chicken. This deletion (1773 bp) contained 27 highly conserved amino acids of the 3′ end of the coding region, the in-frame stop codon, a less frequently used poly(A) signal that is normally found 445 bp downstream of the stop codon, and a large portion of the 3′ UTR. Because of this deletion, 27 novel amino acids were substituted and the open reading frame was extended for an additional 26 amino acids before reaching the transcriptional termination site. The predicted amino acid sequence of the novel carboxyl-terminus of the *dwdw* cGHR is largely hydrophobic with a polylysine tail, whereas the carboxyl-terminus of the wild-type (*DwDw*) cGHR is composed of hydrophilic amino acids. Western blot analysis using antisera directed against the extracellular domain of the cGHR confirmed that the mutant transcript was translated. These observations suggested that the mutation in the intracellular domain results in expression of a dysfunctional cGHR which could account for the phenotype of the CT *dwdw* chicken.

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Introduction

Sex-linked dwarfism of chickens is a genetic disorder characterized by low body weight and long bone growth (Guillaume 1976), and reduced levels of plasma insulin-like growth factor-I (Huybrechts *et al.* 1985) despite normal or high levels of circulating growth hormone (GH) (Scanes *et al.* 1983, Stewart *et al.* 1984). This disorder is analogous to the Laron-type syndrome of humans where several abnormalities of the GH receptor (GHR) gene have been identified (Goossens *et al.* 1993). Sex-linked dwarfism of chickens results from a recessive gene (*dw*) present on the sex (Z) chromosome (Guillaume 1976, Decuypere *et al.* 1992). The sex-linked dwarf (*dwdw*) mutation occurs in different populations of domestic chickens (Guillaume 1976) and, like the Laron-type syndrome, is generated by different mutations in the chicken GHR (cGHR) gene. We have recently reported the absence of hepatic GH-binding activity and abnormal cGHR gene expression in two different strains of *dwdw* broiler chickens (Burnside *et al.* 1992). A restriction-fragment length polymorphism (RFLP) in DNA and an aberrantly sized transcript were found in a strain of *dwdw* chickens maintained at Arbor Acre Farms (Glastonbury, CT, USA). In a second strain maintained at the University of Georgia (Athens, GA, USA), a splice site mutation at the exon 5/intron 5 junction results in the exclusive expression of a small truncated cGHR transcript (Huang *et al.* 1993). A third type of mutation has been identified in a *dwdw* strain of Leghorn chickens where a normal-sized cGHR transcript is detected (Duriez *et al.* 1993). In this *dwdw* strain, a guanosine (G) to thymidine (T) transversion, which generates an isoleucine instead of a serine in the extracellular domain, impairs plasma membrane expression and GH-binding activity. A second strain of *dwdw* Leghorn chicken apparently has a normal-sized cGHR transcript, although the actual defect has not been identified (Hull *et al.* 1993).

The purpose of the present study was to identify the molecular mutation of the cGHR gene of the Connecticut (CT) strain of *dwdw* chicken that results in the production of a 2·6 kb cGHR transcript, instead of the full-length

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4·3 kb transcript found in normal chickens (Burnside *et al.* 1991, 1992). Initially, the normal wild-type (*DwDw*) cGHR gene was mapped and the 3′ untranslated region (3′ UTR) isolated and sequenced. This information was then used to determine the exact mutation in the cGHR gene of the CT strain of *dwdw* broiler chicken. In this *dwdw* strain, we found a large deletion at the 3′ end of the cGHR gene. As a result of this deletion, 27 amino acids of the intracellular domain, the stop codon and a large portion of the 3′ UTR were lost. The aberrant (2·6 kb) cGHR transcript found in liver RNA from the CT strain of *dwdw* chicken was translated as indicated by Western blot analysis.

Materials and Methods

Animals

The strain of *dwdw* chickens used in the present study was maintained at Arbor Acre Farms, Inc. A second strain of *dwdw* chickens was maintained at the University of Georgia. Fertile eggs were shipped to the University of Delaware and hatched. Chicks were maintained in a heated battery brooder until 3 weeks of age, and then transferred to wire cages. White New Zealand male rabbits were used for production of antibody to the extracellular domain of the cGHR. Animals were handled in accordance with the principles and procedures outlined by the University's Animal Care and Use Committee.

Materials

Enzymes were obtained from Promega (Madison, WI, USA) and GIBCO/BRL (Gaithersburg, MD, USA). Sequenase Version 2·0 DNA sequencing kit was purchased from US Biochemicals (USB; Cleveland, OH, USA). Radiolabelled nucleotides were purchased from New England Nuclear (Boston, MA, USA). Oligonucleotides were prepared by Biosynthesis, Inc. (Sanger, TX, USA). Lambda Zap II vector kit for construction of genomic libraries was purchased from Stratagene (La Jolla, CA, USA). Polymerase chain reaction (PCR) products were cloned using the pCR II T-vector from Invitrogen Corporation (San Diego, CA, USA). Superscript RNA PCR kit was purchased from GIBCO/BRL. The pMAL protein fusion and purification system was obtained from New England BioLabs, Inc. (Beverly, MA, USA). The ECL Western blotting system was purchased from Amersham (Arlington Heights, IL, USA).

Genomic DNA isolation and Southern blot analysis

Total chromosomal DNA was isolated using standard procedures (Maniatis *et al.* 1982) and digested with various enzymes. The DNA was separated in a 0·8% agarose gel, denatured, transferred to nylon membrane and u.v. cross-linked. Blots, prewet in 6 × SSC (1 × SSC=0·15 M sodium chloride, 0·015 M sodium citrate, pH 7·0), were pre-hybridized for 4 h at 65 °C in a solution containing 5 × SSC, 10 mM Tris–HCl, pH 7·4, 4 × Denhardt's solution (1 × Denhardt's=0·02% ficoll, 0·02% polyvinyl-pyrrolidone, 0·02% bovine serum albumin), 1% sodium dodecyl sulphate (SDS), 10% dextran sulphate and 100 μg salmon sperm DNA/ml. Probes, from cGHR cDNA (Burnside *et al.* 1991), were random-primer labelled with ^{32}P dCTP, and added directly to the prehybridization solution. Blots were hybridized to the probes for 16–20 h, and then were washed with a final stringency of 0·2 × SSC, 0·1% SDS at 60 °C and exposed to Kodak XAR film at −70 °C with intensifying screens.

Construction and screening of a DwDw *genomic DNA library*

DwDw genomic DNA was digested with EcoRI and size-fractionated through a sucrose gradient to yield 2–10 kb fragments. The fragments were ligated in an average equimolar amount (150 ng) with 1 μg EcoRI-digested and phosphatase-treated Lambda ZAP II arms. The ligation reaction was packaged into Gigapack packaging extract (Stratagene) according to the manufacturer's instructions and the primary recombinants were saved in buffer containing 0·1 M NaCl, 0·01 M $MgSO_4$, 0·05 M Tris–HCl, pH 7·5 and 0·01% (w/v) gelatin. A probe corresponding to nucleotide (nt) 1394 to 2360 of the cGHR cDNA (Burnside *et al.* 1991) was used to screen the genomic library. Conditions for prehybridization, hybridization and washing of duplicate filter lifts were the same as described for the Southern blot analysis. Initially, 10^6 plaques were screened. Plaque purifications of the positive clones obtained from the primary screening and excision of the pBluescript SK$^-$ plasmid carrying the insert from Lambda ZAP II were performed according to the manufacturer's instructions.

Restriction digest analysis of all positive clones showed a 6 kb EcoRI insert. One of the clones was used for further analysis. Beginning at the EcoRI site located in exon 10, about 2·4 kb of sequence was obtained (see Fig. 1). All sequencing was performed using the Sequenase Version 2·0 DNA sequencing kit, according to the manufacturer's instructions.

Identification of the 3′ end of the full-length cGHR transcript from DwDw *chicken*

Poly(A) RNA (1 μg) from *DwDw* chicken was reverse-transcribed with oligo $(dT)_{28}$ using 200 U M-MLV reverse transcriptase according to the instructions for the Superscript cDNA synthesis system (GIBCO/BRL). The cDNA was amplified with primer A (nt 3345 to 3364; see Fig. 2) and oligo $(dT)_{28}$ using PCR conditions as described below. The PCR product (350 bp) was purified from an

agarose gel and cloned into the pCR II T-vector and sequenced as described above.

Isolation of the 3′ end of the dwdw *cGHR gene*

Primer B (nt 1709 to 1728; see Fig. 2) and primer C (nt 3715 to 3734), which flank a 2026 bp region in the 3′ end of *DwDw* cGHR gene, were used to amplify the same region from *dwdw* genomic DNA. PCR was performed with 1 μg genomic DNA in a final reaction volume of 50 μl containing 0·5 μM of each primer, 50 mM KCl, 10 mM Tris–HCl, pH 9·0, 0·1% Triton X-100, 2·5 mM $MgCl_2$, 0·2 mM dNTPs and 1 U Taq DNA polymerase. The amplification conditions were 94 °C for 1 min, 50 °C for 1 min, 72 °C for 2 min, for 35 cycles. A final extension was carried out at 72 °C for 7 min. The PCR product (253 bp) obtained from the *dwdw* genomic DNA was cloned into the pCR II T-vector and sequenced.

Cloning and expression of the extracellular portion of the cGHR

The extracellular portion (without the signal sequence) of the cGHR from *DwDw* chicken ($cGHR_{ex}$) was amplified from the full-length cGHR cDNA clone by PCR with a primer corresponding to nt 90 to 108 and a primer complementary to nt 714 to 738 which is located just before the transmembrane domain. A translational stop codon was included in the downstream primer and restriction enzyme sites, EcoRI and SalI, were included in the upstream and downstream primer respectively. PCR conditions were the same as those described above. The fragment was cloned into corresponding sites of the vector pMAL-c2 (New England Biolabs) and expressed in *Escherichia coli* as a fusion protein (maltose binding protein, MBP–$cGHR_{ex}$). The MBP–$cGHR_{ex}$ fusion protein, located in the cytoplasm, was extracted and purified by affinity chromatography according to the manufacturer's instructions.

Production of polyclonal antibodies to MBP–$cGHR_{ex}$

White New Zealand rabbits were immunized with 1 mg MBP–$cGHR_{ex}$ in complete Freund's adjuvant. They were boosted with 2 mg MBP–$cGHR_{ex}$ in incomplete Freund's adjuvant at 4-week intervals. Blood samples were collected after the second booster immunization and the immunoglobulin G fraction of delipidated serum purified by ammonium sulphate precipitation (50% saturated solution). The recombinant rat GH-binding protein (rGHBP) (kindly provided by Dr William Baumbach, American Cyanamid, Princeton, NJ, USA) was used to characterize the antisera. The anti-MBP–$cGHR_{ex}$ antibody detects the 32 kDa rGHBP (Sadeghi *et al.* 1990) on Western blot analysis.

Western blot analysis

Partially purified liver membranes were prepared by the method described by Krishnan *et al.* (1989) and 100 μg samples electrophoresed according to Laemmli (1970). Proteins were transferred to a nitrocellulose membrane which was then pretreated at 4 °C overnight in phosphate-buffered saline (PBS; pH 7·5) containing 0·1% Tween and 10% milk (PBSTM) and then incubated for 1 h at room temperature in PBSTM containing anti-MBP–$cGHR_{ex}$ antibodies (1:1000 dilution). The blots were washed twice in PBSTM at room temperature and incubated for 1 h with anti-rabbit immunoglobulin G conjugated to horseradish peroxidase (1:2000). Blots were washed again in PBSTM and the bound antibodies were detected with the ECL detection system.

Results

Restriction enzyme map of the cGHR gene

We prepared a restriction enzyme map of the normal cGHR gene from Southern blot analysis of genomic DNA (Fig. 1). Southern blot analysis indicated that there was a single gene which spans approximately 18 kb. The location of the exons was determined by the relative positions of known restriction enzyme sites present in the cGHR cDNA. The map showed three internal NcoI sites, five internal EcoRI sites, six internal HindIII sites, four internal BamHI sites and two internal PstI sites. Exons were numbered according to the human GHR gene (Godowski *et al.* 1989). The intron/exon boundaries were determined for exons 5, 6, 9 and 10 and were identical to those reported for the human GHR. Exon 10 was about 2810 bp and contained most of the intracellular domain; within this exon, there were 1846 bp of 3′ UTR and two functional polyadenylation signals (AATAAA). Two cDNA clones which diverged 11 bases upstream of the initiator methionine have been isolated (unpublished data) indicating that there were two additional exons upstream of exon 2; however, these were not mapped. A chicken homologue of human exon 3 was not found in the cDNA and its presence in the cGHR gene has not been determined.

Location of the deletion in the cGHR gene of dwdw *chicken*

Preliminary experiments using PCR indicated that most of the cGHR coding region was present in the *dwdw* mRNA. However, we have previously reported an RFLP in genomic DNA from *dwdw* chicken using a cDNA probe spanning exons 9 and 10 (Burnside *et al.* 1991, 1992). In normal genomic DNA, this probe hybridizes to a 2·3 kb HindIII fragment (see Fig. 1); however, a 3·2 kb fragment was found in *dwdw* genomic DNA. Based on the HindIII map (Fig. 1), the shift in size of this fragment could be due

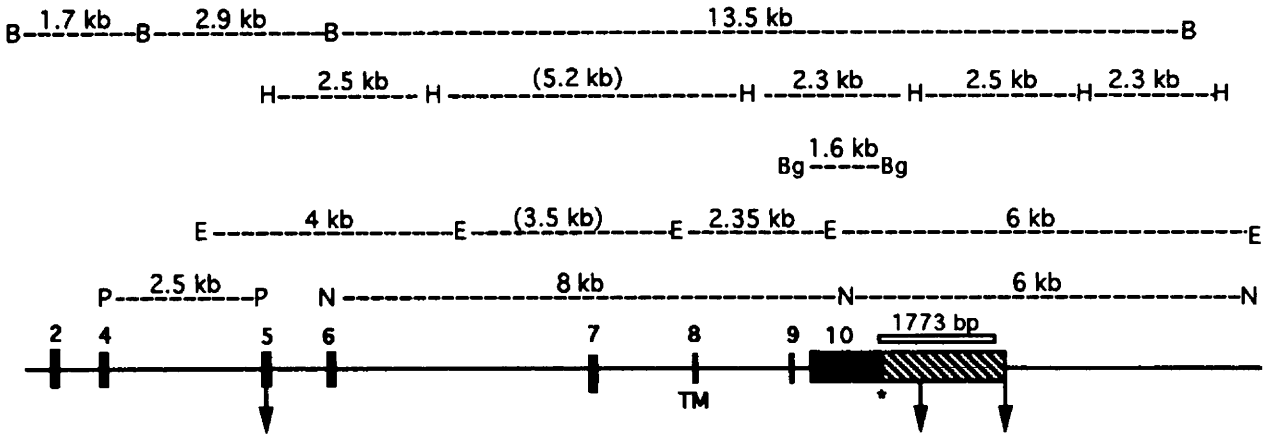


FIGURE 1. Restriction enzyme map of the chicken GH receptor (cGHR) gene. Restriction sites are shown for BamHI (B), HindIII (H), EcoRI (E), BglII (Bg), PstI (P), and NcoI (N). Broken lines show the distance between restriction enzyme sites. Numbers in parentheses indicate presumed fragment size; presence of additional sites within unsequenced introns could not be ruled out. The lengths and numbering of exons, indicated by solid boxes, are based on the structure of the human GHR gene (the presence of a chicken homologue for exon 3 has not been determined). The hatched box represents the 3′ untranslated region and the asterisk indicates the location of the stop codon. Shaded boxes (at 2 and 7) indicate exons for which the exact location could not be determined. The solid lines represent the introns. Arrows indicate the sites of poly(A) signals determined by comparing the cDNA and genomic sequences. The transmembrane domain is indicated by TM. The open box shows the deleted region in the cGHR gene in the CT strain of *dwdw* chicken.

to the loss of the HindIII site within exon 10. An RFLP was also obvious in EcoRI digest of genomic DNA from *dwdw* chickens; a 4·3 kb fragment was found instead of the 6 kb EcoRI fragment seen in *DwDw* genomic DNA. This implied that there was a 1·7 kb deletion in the 3′ region of the gene that included the HindIII site located within exon 10. To confirm this observation, the 3′ end of the cGHR gene was isolated from *DwDw* and *dwdw* genomic DNA.

Cloning of the 3′ end of the cGHR gene from DwDw *chicken*

An EcoRI library of *DwDw* genomic DNA was constructed in Lambda ZAP II to obtain the 6 kb EcoRI fragment which contained the full-length 3′ UTR. The library was screened with a cDNA probe spanning nt 1394 to 2390 of the cGHR cDNA. A clone containing a 6 kb insert was isolated and sequenced. Nucleotide sequence of the complete 3′ UTR of the cGHR is presented in Fig. 2. There were two polyadenylation signals located in the 3′ UTR. The polyadenylation signal at nt 2273 was used to generate the 3·2 kb cGHR transcript found on Northern blots of liver RNA (Burnside *et al.* 1991). The other polyadenylation signal located at nt 3644 was used for synthesis of the full-length (4·3 kb) transcript. To identify the polyadenylation cleavage site utilized to generate the 4·3 kb transcript, poly(A) RNA from *DwDw* chicken was reverse transcribed with oligo (dT). The cDNA was then amplified by PCR using an upstream primer (primer A; Fig. 2) located 1518 bp downstream of the in-frame stop codon and oligo (dT). A polyadenylated PCR product was amplified and sequenced for comparison with the genomic sequence of the corresponding region. The cleavage site for the addition of the poly(A) tail was found 26 bp downstream from the poly(A) signal and is indicated in Fig. 2.

Identification of the deleted region in cGHR of dwdw *chickens*

Two PCR primers (primers B and C; Fig. 2) were designed to flank a 2026 bp region of the *DwDw* cGHR gene. Using *dwdw* genomic DNA as a template, a 253 bp product was amplified; this was 1·7 kb smaller than that amplified from *DwDw* chicken DNA (Fig. 3). After cloning and sequencing the 253 bp fragment, it was apparent that this region of the *dwdw* cGHR lacked 81 nucleotides (coding for 27 amino acids at the carboxyl terminus) and 1692 bp of the 3′ UTR of the wild-type cGHR gene. This deleted region included the translational stop codon and the less frequently used poly(A) signal at nt 2273. The predicted amino acid sequence of the carboxyl-termini of the cGHR of *DwDw* and *dwdw* chickens is shown in Fig. 4. Because of the deletion of the stop codon, 27 novel amino acids were substituted, and the open reading frame was extended for 26 additional amino acids before reaching the terminal polyadenylation cleavage site. The open reading frame of the *dwdw* cGHR could be extended further into the poly(A) tail resulting in a polylysyl-carboxyl terminus. The novel carboxyl tail of the *dwdw* cGHR was predominately hydrophobic whereas the wild-type cGHR had a hydrophilic tail.

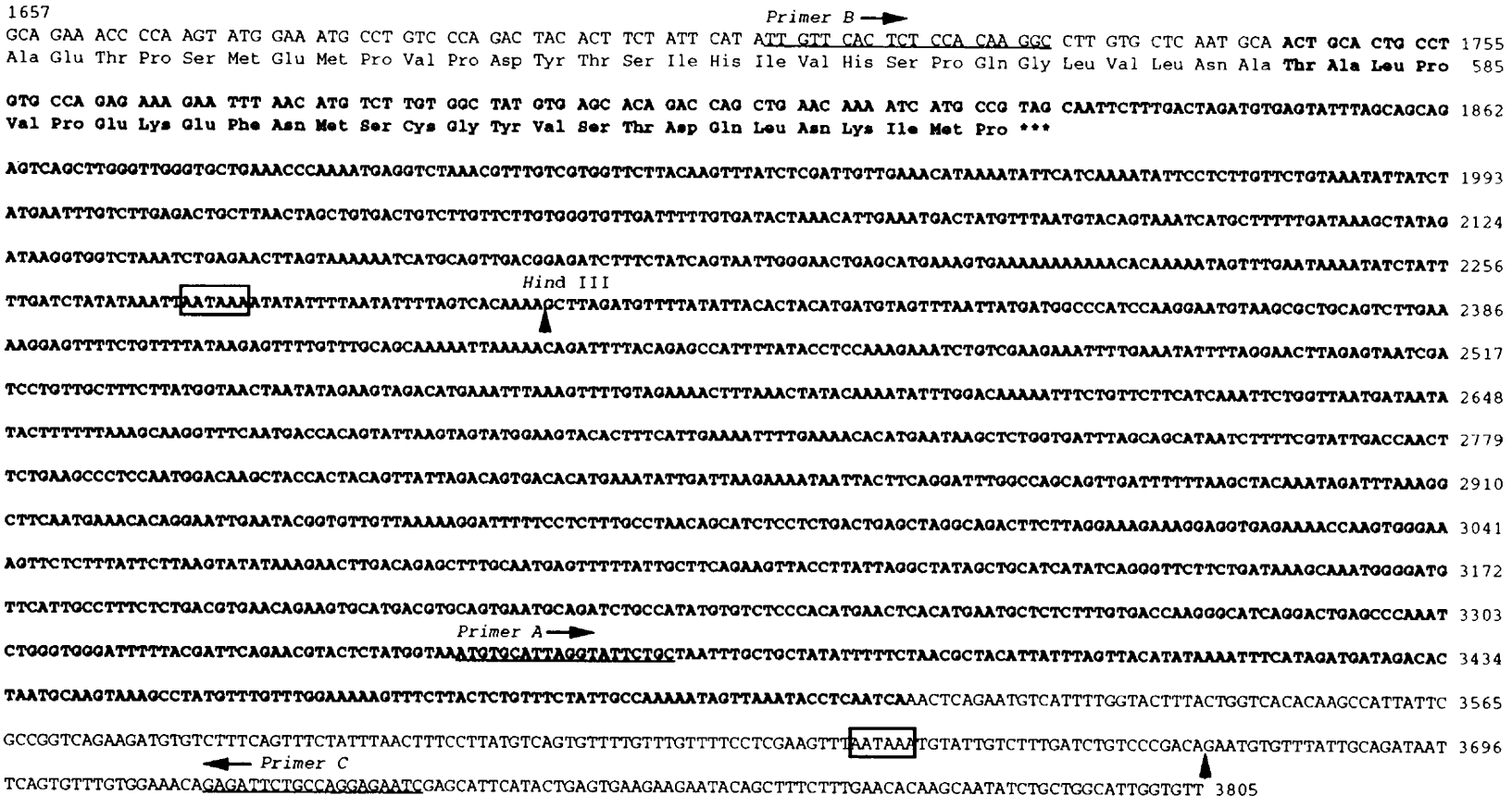


FIGURE 2. Nucleotide sequence of part of the coding region (exon 10) and complete 3′ untranslated region of the chicken GH receptor (cGHR). The sequence shown is from nucleotide 1657, with reference to the adenosine (A) of the initiator methionine in the cDNA sequence as nucleotide 1. Two functional poly(A) signals (AATAAA) are boxed and the polyadenylation cleavage sites are indicated by the upright arrows. Polymerase chain reaction primers A, B and C are indicated by horizontal arrows. Nucleotides 1744 to 3516, shown in bold type, represent the region deleted in the CT strain of *dwdw* chicken.

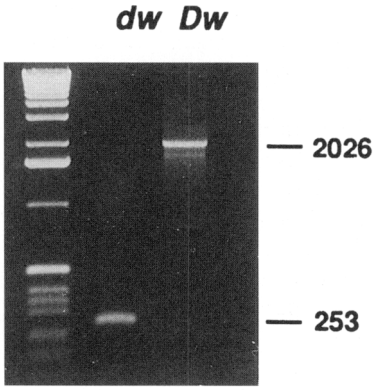


FIGURE 3. Ethidium bromide-stained agarose gel of amplified fragments of wild-type (*Dw*) and dwarf (*dw*) chicken GH receptor gene. Polymerase chain reaction and primers B and C (see Fig. 2) were used to amplify genomic DNA. A 1 kb DNA ladder is shown in the first lane.

Detection of the dwdw *cGHR protein*

A polyclonal antibody against the extracellular portion of the cGHR (anti-$cGHR_{ex}$) was used to determine whether the cGHR mRNA transcribed from the mutated gene of the CT strain of *dwdw* chicken was translated into a protein. Liver membranes were prepared from *DwDw* and *dwdw* chickens of the CT strain and from a different strain of *dwdw* chickens maintained at the University of Georgia (GA). The cGHR of the GA strain of *dwdw* chickens contains a splice site mutation which prevents expression of the full-length cGHR (Huang *et al.* 1993). Western blot analysis showed a broad band around 120 kDa in liver membranes from both *DwDw* and *dwdw* chicken in the CT strain (Fig. 5). As expected, liver membranes from the GA strain of *dwdw* chicken lacked a protein band that corresponds to the cGHR.

Discussion

Analysis of the cGHR gene structure in *dwdw* chickens from the CT strain has revealed a large deletion that includes a portion of the intracellular domain and the 3′ UTR. This finding explains our original observations of the presence of a DNA polymorphism and an aberrantly sized transcript in the *dwdw* chicken (Burnside *et al.* 1991, 1992). Three different transcripts corresponding to the cGHR (0·8, 3·2 and 4·3 kb) have been identified in liver RNA of normal chickens (Burnside *et al.* 1991). The 0·8 kb transcript is generated through the use of the poly(A) signal in exon 5 and lacks a large part of the cGHR coding sequence (Huang *et al.* 1993, Oldham *et al.* 1993). The 3·2 and 4·3 kb transcripts contain the same coding region and are produced as a result of two different poly(A) signals. The 4·3 kb transcript is the most abundant form of the full-length cGHR. In the CT strain of *dwdw* chicken, only two transcripts (0·8 and 2·6 kb) are expressed. The data from the present study indicate that the 2·6 kb transcript results from a simple deletion event at the 3′ end of the cGHR gene. The expression of the 0·8 kb transcript is not affected because it is generated from a poly-adenylation signal embedded in the extracellular domain (Huang *et al.* 1993, Oldham *et al.* 1993). However, the 1·7 kb deletion in the cGHR gene of the CT *dwdw* chicken reduces the size of the predominant transcript to 2·6 kb and eliminates the polyadenylation signal used to generate the 3·2 kb transcript.

In the CT *dwdw* chicken, the deletion of 27 amino acids from the carboxyl-terminus of the normal cGHR is probably responsible for the lack of hepatic GH-binding activity (Burnside *et al.* 1992). Of the 27 amino acids deleted, 18 are conserved in mouse, rat, bovine, human, rabbit and chicken GHRs, including a highly conserved cysteine residue (Leung *et al.* 1987, Baumbach *et al.* 1989, Mathews *et al.* 1989, Smith *et al.* 1989, Adams *et al.* 1990, Hauser *et al.* 1990, Burnside *et al.* 1991). These 27 amino acids in the wild type cGHR are predominantly hydrophilic, and are likely to be exposed on the surface of the protein, allowing them to interact with other cytosolic proteins. The hydrophobic nature of this region in the *dwdw* cGHR could alter folding, interaction with other proteins or the subcellular location. The apparent size of the cGHR detected by Western blot analysis is consistent with the size reported for the rabbit GHR (Spencer *et al.*

```
        573         583         593         603         613         623         633         643
DwDw    HIVHS PQGLVLNATA LPVPEKEFNM SCGYVSTDQL NKIMP*
              ||||| |||||||||
dwdw    HIVHS PQGLVLNANS ECHFGTLLVT QAIIRRSEDV SFSFYLTFLM SVFCLFSSKF NKCIVFDLSR QKKKKKK...
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FIGURE 4. The deduced amino acids of the carboxyl-termini of the *DwDw* and *dwdw* chicken GH receptor (cGHR). Numbers refer to the amino acid sequence of the previously published sequence (Burnside *et al.* 1991). The predicted amino acid sequences are identical through amino acid 581, as indicated by the vertical lines. The 27 amino acids present at the carboxyl-terminus of the *DwDw* cGHR are deleted from the *dwdw* cGHR; instead, 53 novel amino acids and a chain of lysine residues are substituted. The amino acids shown in bold type in the *DwDw* sequence are highly conserved in mammalian and chicken GHRs. The underlined regions of the *dwdw* sequence indicate hydrophobic amino acids.

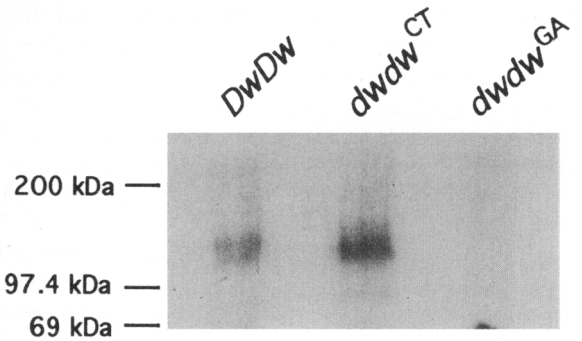


FIGURE 5. Western blot analysis of liver membranes prepared from *DwDw* and *dwdw* chickens. Liver membranes prepared from normal (*DwDw*) chickens and two different strains of *dwdw* chickens ($dwdw^{CT}$ and $dwdw^{GA}$) are shown. Proteins were transferred to a nitrocellulose membrane and sequentially incubated with polyclonal antibody against the extracellular domain of the chicken GH receptor and anti-rabbit immunoglobulin G conjugated to horseradish peroxidase. The complex was detected using chemiluminescence.

1988). However, the resulting protein does not display GH-binding activity (Burnside *et al.* 1992) and, as such, does not participate in signal transduction.

Several studies have addressed the functional role of sequences located in the intracellular domain of the human GHR. Kou *et al.* (1993) have analysed GHR sequence in a patient with Laron syndrome and found two amino acid substitutions in the intracellular domain which resulted in low serum GH-binding activity and growth failure. This indicates that small alterations in the intracellular domain can dramatically affect the function of the GHR. On the other hand, structure/function studies of the GHR have shown that large portions of the intracellular domain can be deleted without altering GH-binding activity or signal transduction. For example, Moldrup *et al.* (1991) found that domains within the carboxyl-terminal half of the intracellular domain of the rat GHR are required for transduction of the signal for GH-stimulated insulin synthesis in rat insulinoma cells. However, deletion of these parts of the intracellular domain does not impair GH-binding activity or translocation to the plasma membrane. In contrast, Colosi *et al.* (1993) have reported that GH-binding activity and signal transduction comparable with that of the full-length GHR were maintained in mutants lacking most of the intracellular domain. It is interesting that not all truncation mutants are expressed, and a truncated mutant lacking 49 carboxyl-terminal amino acids was active, but only poorly expressed. The seemingly conflicting structure/function data could be reconciled if, in fact, different parts of the intracellular domain transduce different signals, as has been found for other members of the haematopoietic/GH receptor superfamily (Fukunaga *et al* 1993). In addition, it seems that modifications of the intracellular domain of mammalian GHRs can have pleiomorphic effects. In the CT *dwdw* chicken, the cGHR mutation does not simply truncate the intracellular domain. The predicted amino acid sequence of the mutated cGHR contains a carboxyl-terminal extension which is predominantly hydrophobic, has two cysteine residues, and a chain of lysine residues encoded by the poly(A) tail. These modifications in the cGHR gene clearly affect GH-binding activity which could result from significant alterations in folding of the protein and/or its subcellular location. Although we have not ruled out the possibility of additional mutations within the *dwdw* gene, it seems reasonable to conclude that alterations in the carboxy-terminus result in the synthesis of a dysfunctional cGHR protein. The results of these studies and others (Duriez *et al.* 1993, Huang *et al.* 1993) indicate that different mutations in the cGHR gene appear to be responsible for the dwarfing phenotype observed in different populations of domestic chickens.

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References

Adams TE, Baker L, Fiddes RJ & Brandon MR 1990 The sheep growth hormone receptor: molecular cloning and ontogeny of mRNA expression in the liver. *Molecular and Cellular Endocrinology* **73** 135–145.

Baumbach WR, Horner DL & Logan JS 1989 The growth hormone-binding protein in rat serum as an alternatively spliced form of the rat growth hormone receptor. *Genes and Development* **3** 1199–1205.

Burnside J, Liou SS & Cogburn LA 1991 Molecular cloning of the chicken growth hormone receptor complementary deoxyribonucleic acid: mutation of the gene in sex-linked dwarf chickens. *Endocrinology* **128** 3183–3192.

Burnside J, Liou SS, Zhong C & Cogburn LA 1992 Abnormal growth hormone receptor gene expression in the sex-linked dwarf chicken. *General and Comparative Endocrinology* **88** 20–28.

Colosi P, Wong K, Leong S & Wood WI 1993 Mutational analysis of the intracellular domain of the human growth hormone receptor. *Journal of Biological Chemistry* **268** 12617–12623.

Decuypere E, Huybrechts LM, Kuhn ER, Tixier-Biochard M & Merat P 1992 Physiological alterations associated with the chicken sex-linked dwarfing gene. *Critical Reviews in Poultry Biology* **3** 191–221.

Duriez B, Sobrier ML, Duquesnoy P, Tixier-Boichard M, Decuypere E, Coquerelle G, Zeman, M, Goossens M & Amselem S 1993 A naturally occurring growth hormone receptor mutation: *in vivo* and *in vitro* evidence for the functional importance of the WS motif

common to all members of the cytokine receptor superfamily. *Molecular Endocrinology* **7** 806–814.

Fukunaga R, Ishizaka-Ikeda E & Nagata S 1993 Growth and differentiation signals mediated by different regions in the cytoplasmic domain of granulocyte colony-stimulating factor receptor. *Cell* **74** 1079–1087.

Godowski PJ, Leung PW, Meacham LR, Galagani JP, Hellmiss R, Keret R, Rotwein PS, Parks JS, Laron Z & Wood WI 1989 Characterization of the human growth hormone receptor gene and demonstration of partial gene deletion in two patients with Laron-type dwarfism. *Proceedings of the National Academy of Sciences of the USA* **86** 8083–8087.

Goossens M, Amselem S, Duquesnoy P & Sobrier M-L 1993 Molecular genetics of Laron-type GH insensitivity syndrome. *Recent Progress in Hormone Research* **48** 165–178.

Guillaume J 1976 The dwarfing gene dw: Its effect on anatomy, physiology, nutrition and management. *World Poultry Science Journal* **32** 285–304.

Hauser SD, McGrath MF, Collier RJ & Krivi GG 1990 Cloning and *in vivo* expression of bovine growth hormone receptor mRNA. *Molecular and Cellular Endocrinology* **72** 187–200.

Huang H, Cogburn LA, Agarwal SA, Marks HL & Burnside J 1993 Over-expression of a truncated growth hormone receptor in the sex-linked dwarf chicken: Evidence for a splice mutation. *Molecular Endocrinology* **7** 1391–1398.

Hull KL, Fraser RA, Marsh JA & Harvey S 1993 Growth hormone receptor gene expression in sex-linked dwarf leghorn chickens: evidence against a gene deletion. *Journal of Endocrinology* **137** 91–98.

Huybrechts LM, King DB, Lauterio TJ, Marsh J & Scanes CG 1985 Plasma concentrations of somatomedin-C in hypophysectomized, dwarf and intact growing domestic fowl as determined by heterologous radioimmunoassay. *Journal of Endocrinology* **104** 233–239.

Kou K, Lajara R & Rotwein P 1993 Amino acid substitution in the intracelluar part of the growth hormone receptor in a patient with the Laron syndrome. *Journal of Clinical Endocrinology and Metabolism* **76** 54–59.

Krishnan KA, Proudman JA & Bahr JM 1989 Avian growth hormone receptor assay: Use of chicken and turkey membranes. *Molecular and Cellular Endocrinology* **66** 125–134.

Laemmli UK 1970 Cleavage of structural proteins during the assembly of the head of the bacteriophage T4. *Nature* **227** 680–685.

Leung DW, Spencer SA, Cachianes G, Hammonds RG, Collins C, Henzel WJ, Barnard R, Waters MJ & Wood WI 1987 Growth hormone receptor and serum binding protein: purification, cloning and expression. *Nature* **330** 537–543.

Maniatis T, Fritsch EF & Sambrook J 1982 *Molecular Cloning. A Laboratory Manual.* Cold Spring Harbor Laboratory: Cold Spring Harbor.

Mathews LS, Enberg B & Norstedt G 1989 Regulation of rat growth hormone receptor gene expression. *Journal of Biological Chemistry* **264** 9905–9910.

Moldrup A, Allevatom G, Dyrberg T, Nielsen JH & Billestrup N 1991 Growth hormone action in rat insulinoma cells expressing truncated growth hormone receptors. *Journal of Biological Chemistry* **266** 17441–17445.

Oldham ER, Bingham B & Baumbach WR 1993 A functional polyadenylation signal is embedded in the coding region of the chicken growth hormone receptor RNA. *Molecular Endocrinology* **7** 1379–1390.

Sadeghi H, Wang BS, Lumanglas AL, Logan JS & Baumbach WR 1990 Identification of the origin of the growth hormone-binding protein in rat serum. *Molecular Endocrinology* **4** 1799–1805.

Scanes CG, Marsh J, Decuypere E & Rudas P 1983 Abnormalities in the plasma concentrations of thyroxine, tri-iodothyronine and growth hormone in sex-linked dwarf and autosomal dwarf White Leghorn domestic fowl (*Gallus domesticus*). *Journal of Endocrinology* **97** 127–135.

Smith WC, Kuniyoshi J & Talamantes F 1989 Mouse serum growth hormone (GH) binding protein has GH receptor extracellular and substituted transmembrane domains. *Molecular Endocrinology* **3** 984–990.

Spencer SA, Hammonds G, Henzel WJ, Rodriguez H, Waters MJ & Wood WI 1988 Rabbit liver growth hormone receptor and serum binding protein. *Journal of Biological Chemistry* **263** 7862–7867.

Stewart PA, Washburn KW & Marks HL 1984 Effect of the *dw* gene on growth, plasma hormone concentrations and hepatic enzyme activity in a randombred population of chickens. *Growth* **48** 59–73.

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